

TRANSFORMER PRUNING: FROM LOCAL SALIENCY CRITERIA TO GLOBAL CONFIGURATION SEARCH

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AIMS

This poster surveys the state of the art in pruning Transformer models and organises it along three axes:

- **What is removed** — the sparsity topology and the structural unit.
- **What drives removal** — the saliency criterion that estimates a unit’s importance.
- **How removal is scheduled** — the pruning strategy and the budget allocation across layers.

The recurring limitation of *local*, per-unit scoring motivates a shift toward *global*, *budget-constrained configuration search*.

WHY PRUNE TRANSFORMERS?

Transformer quality scales with size, but deployment is bounded by memory, bandwidth, latency, and energy. Pruning reduces the active parameters or executable units while preserving predictive behaviour, exploiting the pronounced over-parameterisation of modern models.

Model	Params	FP16 (GB)	Type
ResNet-50	25.6 M	0.05	Vision
BERT-Base	110 M	0.22	Encoder LM
GPT-2 (XL)	1.56 B	3.12	Decoder LM
LLaMA-2 (7B)	6.74 B	13.5	Open LLM
LLaMA-2 (70B)	68.9 B	138	Multi-GPU

Table 1:Parameter counts and FP16 memory footprints. Even at 50% sparsity a 70B model still needs ~ 70 GB, so practical gains require removal patterns the hardware can exploit.

THE PRUNING PROBLEM

A binary mask M selects which parameters of a trained network $f(x; \Theta)$ remain active. Pruning is the constrained problem

$$\min_{\Theta, M} \mathcal{L}(M \odot \Theta) \quad \text{s.t.} \quad \frac{\|M\|_0}{P} \leq \kappa,$$

with target density κ . Finding the optimal mask is an NP-hard search over $\{0, 1\}^P$; practical methods rely on saliency surrogates, differentiable relaxations, or greedy heuristics.

RESULTS: UNSTRUCTURED PRUNING

WikiText perplexity ($\downarrow$) at 50% sparsity, comparing magnitude, second-order (SparseGPT), and activation-aware (Wanda) criteria.

Model	Set.	Mag.	SpGPT	Wanda
LLaMA-7B	50%	17.29	7.22	7.26
LLaMA-65B	50%	5.90	4.57	4.57
OPT-13B	50%	1.2e4	11.17	–
OPT-13B	2:4	–	12.96	–

Table 2:Magnitude pruning collapses at 50% (OPT-13B dense PPL is 10.13); second-order and activation-aware criteria stay close to dense.

THEORETICAL FRAMEWORK

Expanding the loss around the trained weights Θ^* gives, with $\delta\Theta = M \odot \Theta^* - \Theta^*$,

$$\Delta\mathcal{L} \approx \nabla\mathcal{L}^\top \delta\Theta + \frac{1}{2} \delta\Theta^\top H \delta\Theta.$$

At convergence the gradient term vanishes and curvature dominates. This single expansion generates the whole family of criteria.

Order	Family	Proxy
0th	Magnitude	$ \theta_i $
1st	Gradient	$ \partial_\theta \mathcal{L} \cdot \theta_i $
2nd (diag)	OBD	$\frac{1}{2} H_{ii} \theta_i^2$
2nd (block)	OBS / SparseGPT local re-fit	δW

Table 3:Criterion families from the Taylor expansion. OBS attains lower loss increase than OBD but needs a per-layer Hessian inverse.

The Lottery Ticket Hypothesis adds that sparse sub-networks can match the dense model, so sparsity is compatible with full accuracy.

PRUNING STRUCTURES

The mask granularity sets both the optimisation freedom and the achievable speedup.

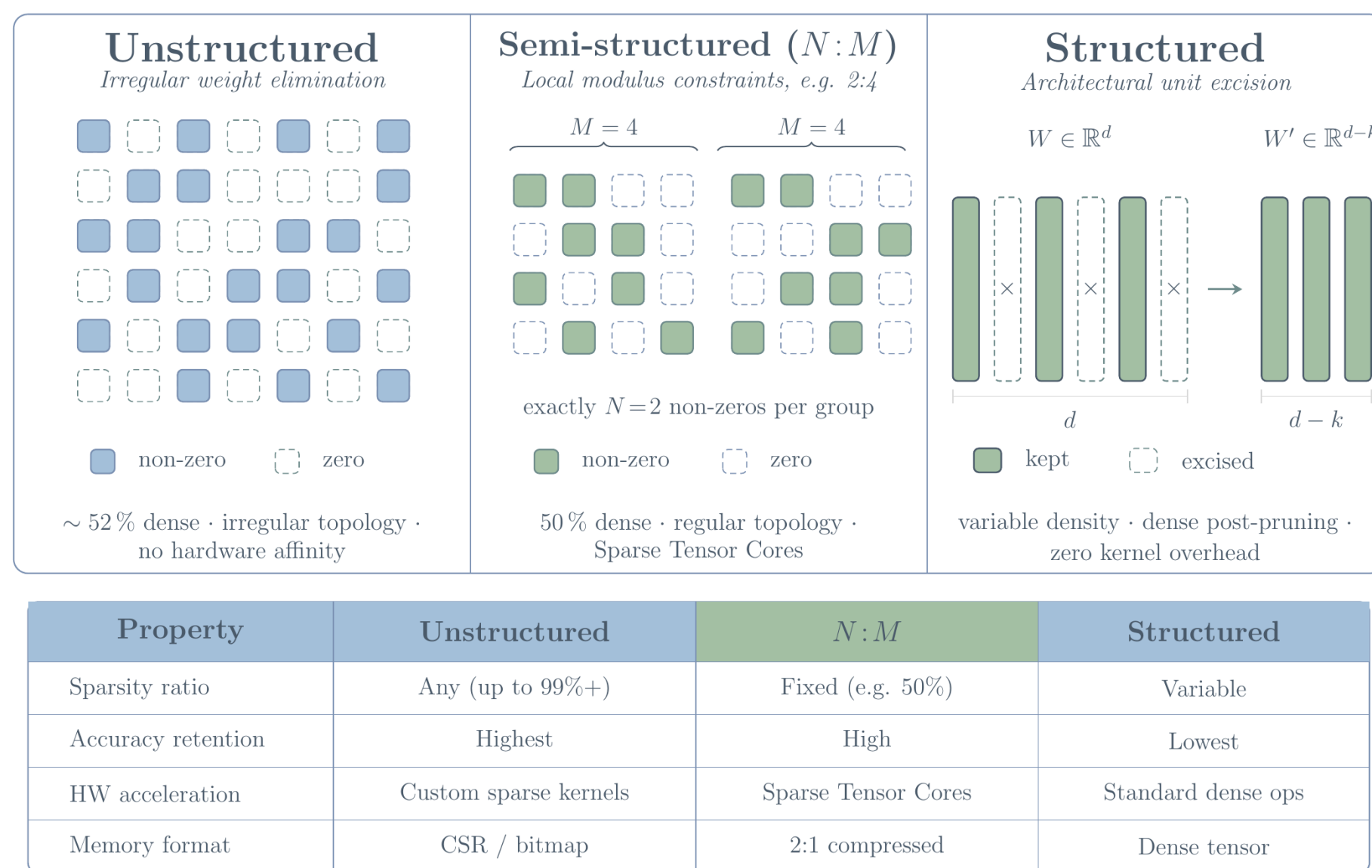


Figure 1:Sparsity topologies: unstructured, semi-structured ($N:M$), and structured.

Only **structured** removal yields a smaller *dense* network that runs fast on standard kernels; unstructured zeros need sparse engines, and 2 : 4 semi-structured sits in between (Ampere tensor cores).

Arch.	Unit	Effect
MLP	Neuron	Hidden width
CNN	Filter	Feature maps
Transf. Head		Attention diversity
Transf. FFN channel		Intermediate width
Transf. Layer		Depth

Table 4:Structured units by architecture.

THE CENTRAL TENSION

Almost every criterion scores each unit **locally and independently**, before the effect of the full set of removals is known. At high sparsity this breaks down: a unit that looks redundant in isolation may become essential once its neighbours are gone, and the way the budget is *allocated across layers* influences final quality more than any single local score. Sparsity alone does not characterise a pruned network — the configuration does.

PRUNING STRATEGIES

Strategy choice (*when, how often, in what order* the mask is updated) is largely orthogonal to criterion choice: **one-shot** (cheap, errors compound), **iterative + recovery** (prune–finetune rounds on a schedule), **during training** (dynamic sparse training, ℓ_1), and **post-training** (frozen model, calibration set only).

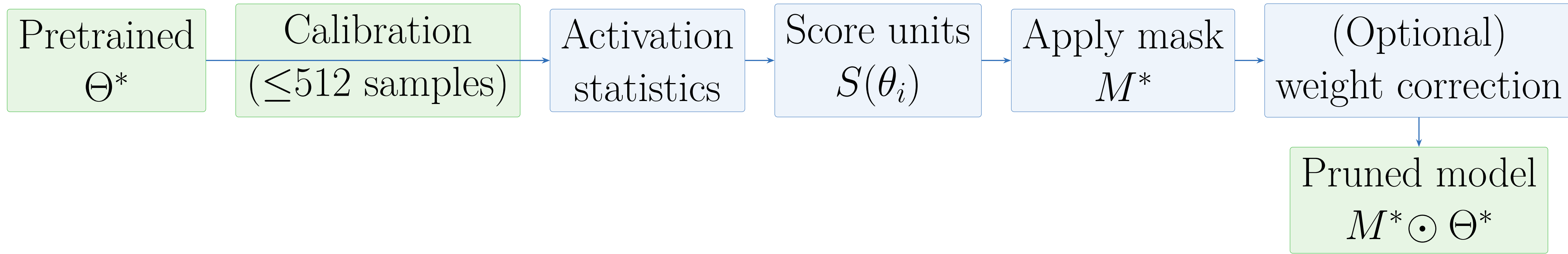


Figure 2:Post-training pruning pipeline: no full retraining, only calibration statistics and optional local weight correction (SparseGPT).

PRUNING CRITERIA

A criterion assigns an importance score as a proxy for the loss change of removing a unit.

Criterion	Signal	Acc.
Magnitude	weights only	base
Gradient	$\text{grad} \times \text{weight}$	+1–3%
OBD	diag. curvature	+1–5%
SparseGPT	per-layer H^{-1}	best PT
Wanda	$\ W\ \cdot \ x\ $	strong PT
Movement	score trajectory	best FT

Table 5:Criteria differ in the information used and the data/compute required; gains are relative to magnitude pruning (PT = post-training, FT = fine-tune).

PRUNING TRANSFORMERS

The removable units are not simple neurons: they are attention heads, FFN channels, whole layers, and (dynamically) tokens.

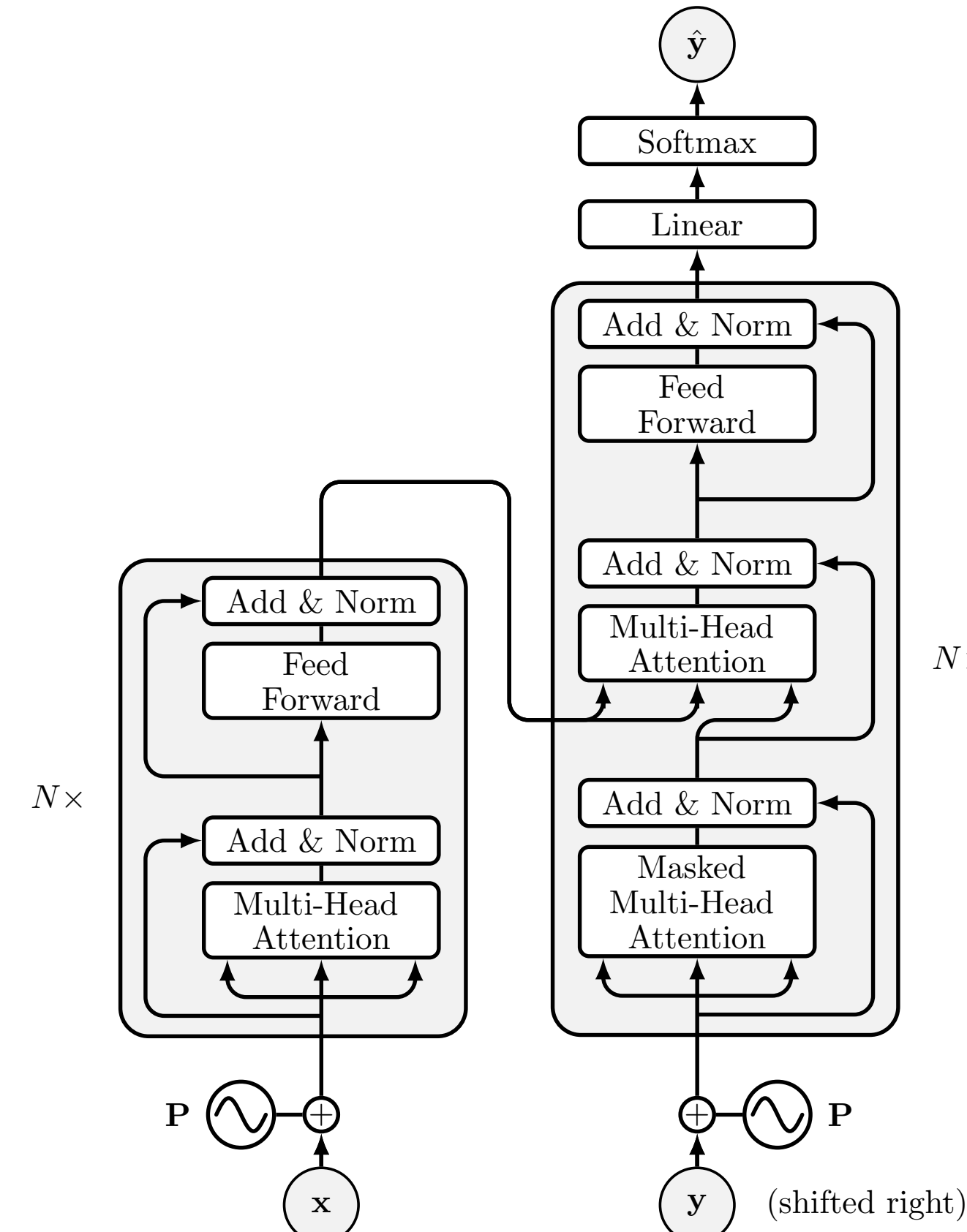


Figure 4:Encoder–decoder Transformer. Heads, FFN channels, and layers are the structured pruning targets; tokens are pruned per input.

- **Head:** removes 4 matrices per head, $\Delta P = 4d^2/H$; 20–40% of heads are often removable.
- **FFN channel:** masks rows of W_1 and columns of W_2 ; entangled under SwiGLU.
- **Layer:** drops a residual block by its contribution $\|F_i(X)\| / \|X\|$; cheap depth, costly capacity.
- **Token:** input-dependent; targets the $O(T^2)$ attention cost.

RESULTS: SEARCH AT HIGH COMPRESSION

WikiText-2 / C4 perplexity ($\downarrow$). Global allocation search (EvoPress) versus block-scoring depth pruning (ShortGPT) and uniform allocation.

Model / Budget	Method	Wiki2	C4
Mistral-7B, 25%	EvoPress	8.66	12.04
	ShortGPT	43.26	40.16
Llama-2-7B, 37.5%	EvoPress	17.98	18.91
	ShortGPT	70.94	63.51
Mistral-7B, 70%	EvoPress	14.42	16.46
	Uniform	23.08	30.03

Table 6:At aggressive budgets, global configuration search dominates local block scoring and uniform allocation by large margins.

DISCUSSION

- **Local vs. global.** Criteria such as SparseGPT and Wanda score units *before* the compressed model is fixed; they identify removable weights but do not decide how to distribute the budget across layers.
- **High-compression limit.** Local ranking assumes removals are independent, but residual and attention structure couple layers — so allocation search (ZipLM, EvoPress) becomes the dominant factor.
- **Comparability.** Reported gains span five evaluation regimes and very different recovery budgets (Fisher ≈ 39 s vs. CoFi 40 epochs); post-training methods are also calibration-sensitive.

KEY REFERENCES

- LeCun et al., *Optimal Brain Damage*, 1989.
- Hassibi & Stork, *Optimal Brain Surgeon*, 1993.
- Frankle & Carbin, *Lottery Ticket Hypothesis*, 2019.